

LM02 Analyzing Income Statements

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1. Introduction

The income statement presents information on the financial results of a company's business activities over a period of time. It is also known as the 'statement of operations', 'statement of earnings', or 'profit and loss (P&L) statement'. The basic equation underlying the income statement is:

$$\text{Income} - \text{Expenses} = \text{Net Income}$$

Equity analysts carefully analyze a company's income statements for use in valuation models while fixed-income analysts analyze income statements to measure a company's debt servicing ability.

Components of the income statement

The components of an income statement are:

Revenues: Income generated from the sale of goods and services in the normal course of the business. Net revenue is the total revenue minus products that were returned and amounts that are unlikely to be collected.

Expenses: Costs incurred to generate revenues. Expenses may be grouped and reported in different formats, subject to some specific requirements.

Gains and losses: Amounts generated from non-operating activities.

Net income: Net income can be calculated as $\text{Net income} = \text{Revenues} - \text{Expenses} + \text{Gains} - \text{Losses}$.

A sample income statement is shown below:

Multi-step format		
\$ million	2018	2017
Sales	35,310	31,600
Cost of sales	10,300	9,060
Gross Profit	25,010	22,540
Gain from sale of equipment	900	860
Administrative expenses	3,400	2,900
Advertising expense	1,000	900
Depreciation	960	850
Other expenses	6,500	6,100
Operating Income (EBIT)	14,050	12,650
Interest Expense	10	70
Profit before tax (EBT)	14,040	12,580
Tax Expense	3,945	3,300
Profit after tax	10,095	9,280

2. Revenue Recognition

General Principles

Under the accrual method of accounting, revenue should be recognized when earned and not necessarily when cash is received. Let us consider three simple examples to illustrate this point.

- If a company sells goods for \$100 cash in Period 1, can it recognize revenue in Period 1? The answer is yes. Revenue is recorded in the period it is earned, i.e., when goods or services are delivered.
- What if the company sells goods on credit in Period 1 and expects to receive cash in Period 2? Can revenue be recognized in Period 1? The answer is that revenue is recorded in Period 1. In addition, since the goods are sold on credit, an asset called accounts receivable is created.
- What if an advance payment is received in Period 1 but goods and services are to be delivered in Period 2. When will the revenue be recognized? The revenue will be recognized in Period 2 because that is when delivery of goods will take place. In this case, the company will record a liability called unearned revenue when the advance payment is received.

Companies must disclose their revenue recognition policies in the notes to their financial statements, and analysts should read these carefully to understand how and when a company recognizes revenue.

Accounting Standards for Revenue Recognition

In May 2014, the IASB and FASB issued converged standards for revenue recognition. The standards take a principles-based approach to revenue recognition issues. The core principle behind the converged standard is that revenue should be recognized to “depict the transfer of promised goods or services to customers in an amount that reflects the consideration to which the entity expects to be entitled in an exchange for those goods or services.”

According to the standard, the following five steps must be followed in order to recognize revenue:

1. Identify the contract(s) with a customer.
2. Identify the performance obligations in the contract.
3. Determine the transaction price.
4. Allocate the transaction price to the performance obligations in the contract.
5. Recognize revenue when (or as) the entity satisfies a performance obligation.

When revenue is recognized, a contract asset is added to the balance sheet. If an advance is received but performance obligations have not been met, then a contract liability is added to the seller's balance sheet.

The entity will recognize revenue when it is able to satisfy performance obligation by

transferring control of the goods or service to the customer. The following factors can be considered to determine whether the customer has gained control:

- Entity has a present right to payment
- Customer has legal title
- Customer has physical possession
- Customer has the significant risks and rewards of ownership
- Customer has accepted the good or service

Analysts can encounter many companies with complex revenue recognition policies. Several examples adapted from real companies are presented in the example below.

Instructor's Note: Understand the core concepts covered in this example. On the exam you will probably be tested on a short mini-scenario resembling the scenarios below.

Example: Applying the Converged Revenue Recognition Standards

(This is Example 1 from the curriculum.)

Principal Versus Agent

MegaDigital is an online marketplace that sells goods and delivers them quickly to customers. For some sales, MegaDigital acts as a principal in which it controls the product before the goods are transferred to the customer. In other sales, MegaDigital acts as an agent in which it arranges for the transfer of a product controlled by a third-party seller. In transactions in which MegaDigital is the principal, revenue is recorded as the total amount of considerations received for the transfer of the product. In transactions in which MegaDigital is the agent, it records revenue only for the portion of the considerations, which amounts to its fee or commission. This can have a significant impact on common size and ratio analysis. Revenue is lower but profit margins are higher for sales for which MegaDigital is an agent.

Assume MegaDigital sells a particular product as a principal for USD100 that it purchased for USD70. Additionally, there are USD10 of other selling, general, and administrative costs. The margins would be:

Sales	USD100	100%
Cost of Sales	70	70%
Gross Profit	30	30%
SG&A	10	10%
Net Profit	20	20%

If MegaDigital acts an agent for the same item with the same retail price, MegaDigital would receive a commission of USD30 and still incurs USD10 of other costs. Margins would be:

Sales	USD30	100%
Cost of Sales	0	0%

Gross Profit	30	100%
SG&A	10	33%
Net Profit	20	67%

For companies selling both as a principal and agent, such as many e-commerce companies, an analyst would need to evaluate the relative proportion of principal versus agent sales to evaluate and forecast overall margins. This is especially important if the mix of principal and agent sales is expected to change.

Franchising/Licensing

Mahjong Pizza both operates and franchises pizza delivery restaurants around the world. Revenue recognition standards require that the company disaggregate revenue from contracts with customers into categories that depict how the nature, amount, timing, and uncertainty of revenue and cash flows are affected by economic factors. Companies must present revenues disaggregated in consolidated statements of income to satisfy this requirement. Mahjong Pizza presents the following disaggregated revenue items:

- company-owned stores revenues,
- franchise royalties and fees, and
- supply chain revenues.

Company-owned stores revenues are of retail sales of food at stores that Mahjong owns and operates.

Franchise royalties and fees are comprised of fees from third-party franchisees that are licensed to operate Mahjong restaurants. Each franchisee is generally required to pay fees equal to 5.5 percent of restaurant sales. The company recognizes the royalty fee as revenue, not the total sales of the franchisees' restaurants.

Upfront fees for opening new units are initially recognized as deferred revenue and subsequently amortized to revenue on a straight-line basis over the term of each respective franchise agreement, typically 10 years.

Supply chain revenues are primarily composed of sales of food, equipment, and supplies to franchisees. Revenues are recognized upon delivery or shipment of the related products to franchisees, based on shipping terms.

Software as a Service or License

CReaM Software and Services is a technology company providing customer relationship management software and services to business, government and not-for-profit organizations. Organizations may purchase a software license and install it on their own systems. Alternatively, they may subscribe to CReaM's cloud services platform through

which they can access CReaM's software over the internet for a monthly subscription fee.

Under IFRS 15, if a company provides a license to use software where the company will take possession of the software for installation on their own system, the company will report revenue either over the term of the license or at the time of the transfer of the license.

Companies should report the revenue from the license over the term of the license, if under the contract or the company's normal business activities:

- the software provider will continue to undertake activities that significantly affect the software (e.g., upgrades/enhancements)
- the rights expose the customer to positive or negative impacts from those activities, and
- the activities do not result in a transfer of goods or services.

If these criteria are not met, then the revenue is recognized when the license is transferred to the customer. CReaM's annual report footnotes state:

"Software revenues include revenues associated with term and perpetual software licenses that provide the customer with a right to use the software as it exists when made available. Revenues from term and perpetual software licenses are generally recognized at the point in time when the software is made available to the customer. Revenue from software support and updates is recognized as the support and updates are provided, which is generally ratably over the contract term."

Under the terms of CReaM's license, the software is sold "as is" and revenue is recognized at the time of the license transfer. CReaM, however, also provides a support contract for updates for which revenue is recognized over the contract term.

CReaM's cloud clients have access to constantly updated software. CReaM reports:

"Cloud services allow customers to use the Company's software without taking possession of the software. Revenue is generally recognized over the contract term. Substantially all of the Company's subscription service arrangements are non-cancelable and do not contain refund-type provisions."

In the case of CReaM, an analyst must understand the composition of revenue between licensed software in which case revenue is recognized upfront versus software as a service in which case revenue is recognized over time.

Long-Term Contracts

Armored Vehicles Inc. (AVI) manufactures weapons systems and vehicles for military customers. The company enters long-term contracts that generally extend over several years. Performance on the contracts is satisfied over time. Under IFRS 15, a performance obligation is satisfied over time if one of the following criteria is met:

- The customer simultaneously receives and consumes the benefits provided by the entity's performance as the entity performs (e.g., routine service contracts).
- The entity's performance creates or enhances an asset that the customer controls as the asset is created or enhanced (e.g., refurbishment of a factory owned and controlled by the customer or building a road for a governmental agency).
- The entity's performance does not create an asset with alternative use to the entity and the entity has an enforceable right to payment for performance completed to date (e.g., construction of a large unique asset that may not be able to be sold to another customer such as a weapons system).

AVI recognizes long-term contract revenue over the contract term as the work progresses, either as products are produced or as services are rendered because of the continuous transfer of control to the customer. For its military contracts, this continuous transfer of control to the customer is supported by clauses in the contract that allow the customer to unilaterally terminate the contract for convenience, pay for costs incurred plus a reasonable profit, and take control of any work in process.

Under IFRS 15, the extent of progress towards completion may be measured by output methods (e.g., appraisals or units completed) or input methods (e.g., costs incurred relative to estimated total costs). AVI reports that its accounting for long-term contracts involves a judgmental process of estimating total sales, costs and profit for each performance obligation. Cost of sales is recognized as incurred. The amount reported as revenues is determined by adding a proportionate amount of the estimated profit to the amount reported as cost of sales. Recognizing revenue as costs are incurred provides an objective measure of progress on the long-term contract and thereby best depicts the extent of transfer of control to the customer.

As an example, AVI has a contract to produce a weapons system for a total price of USD10 million. The expected total costs to produce the system is USD7 million and the estimated profit is USD3 million. The system will take two years to produce. In Year 1 of the contract, AVI incurs USD4.2 million of costs representing 60 percent of total estimated costs. AVI would recognize revenue of USD6 million and profit of USD1.8 million in Year 1 (both 60 percent of expected revenue and profits).

If in Year 2, the system is completed with actual total cumulative costs of USD7.5 million, the company would report revenue of USD4 million and costs of USD3.3 million for a Year 2 profit of USD0.7 million and cumulative profit of USD2.5 million.

Bill and Hold Arrangements

In addition to the long-term contracts discussed previously, AVI produces custom armored vehicles that some customers may not be able to take possession of immediately (because,

for example, a lack of storage space). IFRS 15 provides that in such a “bill and hold” arrangement AVI can determine when it has satisfied its performance obligation based on when a customer obtains control of the product. Under IFRS 15, this is when all the following criteria are met:

- The reason for the bill and hold arrangement must be substantive (e.g., the customer has requested the arrangement).
- The product must be identified separately as belonging to the customer.
- The product currently must be ready for physical transfer to the customer.
- The entity cannot have the ability to use the product or to direct it to another customer.

In AVI’s case, each vehicle is identified by a unique vehicle identification number and upon completion, title and risk of loss has passed to the customer. AVI recognizes revenue when the product is ready for delivery to the customer but is directed by the customer to hold delivery.

Disclosure requirements

The converged standard mandates the following disclosure requirements:

- Companies must disclose information about contracts with customers after segregating them into different categories of contracts. The categories may be based on the geographic region, the type of product, the type of customer, pricing terms, etc.
- Companies must disclose information related to revenue recognition. For example, any change in judgments, remaining performance obligations, and transaction price allotted to those obligations, and balances of contract-related assets and liabilities.

3. Expense Recognition

Expenses are ‘decreases in economic benefit during the accounting period in the form of outflows or depletion of assets or incurrences of liabilities that result in decreases in equity.’ For example, if a company pays rent, its cash reduces and the rent is recognized as an expense.

General Principles

Matching principle

The most important principle of expense recognition is the matching principle, under which the expenses incurred to generate revenue are recognized in the same period as revenue. For example, if some goods bought in the current year remain unsold at the end of the year, they are not included in the cost of goods sold for the current year. If they are sold in the next year, they will be included in the cost of goods sold for the next year.

Periodic costs

Expenses that cannot be tied directly to generation of revenues are called periodic costs. They are expensed in the period incurred. For example, the rent paid for office premises are simply expensed in the period for which the rent was paid.

Capitalization versus Expensing

Capitalizing: In general, when a company acquires a long-lived tangible or intangible asset, its cost is capitalized if the asset is expected to provide economic benefits beyond a year. The company records an asset in an amount equal to the acquisition cost plus any other cost to get the asset ready for its intended use.

Capitalizing results in spreading the cost of acquiring an asset over a specified period of time instead of immediately expensing it. All other costs to make the asset ready for intended use are also capitalized. Capitalizing leads to higher profitability in the period when the asset is purchased. The effect of capitalizing an expenditure on the financial statements is summarized below:

<i>Effect of capitalization on financial statements</i>	
Initially when an expenditure is capitalized.	Balance sheet: non-current assets increase by the capitalized amount.
	Statement of cash flows: investing cash flow decreases.
Subsequent periods over the asset's useful life.	Income statement: depreciation or amortization expense. Net income decreases.
	Balance sheet: non-current assets (carrying value of the asset) decreases.
	Retained earnings decreases. Equity decreases.

Expensing: The cost of an asset is expensed if it has uncertain or no impact on future earnings and provides economic benefit only in the current period. Immediate recognition of an asset's cost as an expense on the income statement results in lower profitability in the current period and higher profits in the future.

<i>Effect of expensing on financial statements</i>	
When an expenditure is expensed.	Income statement: Net income decreases by the after-tax amount of the expenditure.
	No depreciation/amortization expense.
	Balance sheet: No asset is recorded.
	Lower retained earnings due to lower net income.
	Statement of cash flow: Operating cash flow decreases.

The table below summarizes the effects of capitalizing versus expensing on various financial statement items.

	<i>Capitalizing</i>	<i>Expensing</i>
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Total assets	Higher	Lower
Equity	Higher	Lower
Income variability	Lower	Higher
Net income (1 st year)	Higher	Lower
Net income (later)	Lower	Higher
CFO	Higher	Lower
CFI	Lower	Higher
D/E	Lower	Higher
Interest coverage (initially)	Higher	Lower
Interest coverage (later)	Lower	Higher
ROA and ROE (initially)	Higher	Lower
ROA and ROE (later)	Lower	Higher

Capitalization of Interest Costs

When an asset requires a long period of time to get ready for its intended use, the interest costs associated with constructing or acquiring the asset are capitalized.

If the interest expenditure is incurred in connection with constructing an asset for the company's own use, capitalized interest is reported as part of the asset's cost on the balance sheet; in the future, it is reported as part of the asset's depreciation expense in the income statement.

If the interest expense is incurred in connection with the construction of an asset to be sold, such as by a real estate construction company, the capitalized interest is recorded on the company's balance sheet as inventory. When the asset is sold, the capitalized interest is expensed as part of the cost of sales.

Effect of Capitalized Interest on Financial Statements:

- Higher net income and greater interest coverage ratios during the period of capitalization.
- Higher asset values and depreciation lead to lower net income, EBIT and interest coverage ratio in the subsequent periods.

To provide a true picture of a company's interest coverage, interest coverage ratios should be calculated using the entire amount of interest expenditure, including both the capitalized and expensed portions.

Also, if a company is depreciating interest that was capitalized in a previous period, income should be adjusted to remove the effect of that depreciation.

Capitalization of Internal Development Costs

Accounting standards require companies to capitalize software development costs after a product's feasibility is established. Because determining feasibility requires judgment,

capitalization practices may differ between companies.

Effect on Financial Statements:

- Income statement - Expensing rather than capitalizing development costs results in lower net income in the current period.
- Cash flow statement – Expensing rather than capitalizing development costs results in lower net operating cash flows and higher net investing cash flows.

While comparing a company that mostly expenses its software development costs with a company that mostly capitalizes its software development costs, analysts can make the following adjustments.

For the company that capitalizes:

- Adjust the income statement to include software development costs as an expense and exclude amortization of prior year's software development costs.
- Adjust the balance sheet to exclude capitalized software. This will decrease assets and equity.
- Adjust the cash flow statement to decrease operating cash flows and decrease cash used in investing by the amount of the current period development costs.

Implications for Financial Analysts: Expense Recognition

If a company's policies result in early recognition of expenses, it can be considered a conservative approach. On the other hand, if a company's policies delay the recognition of expenses, it can be considered an aggressive approach. Using this as well as other information contained in the footnotes or disclosures, an analyst can recognize whether a company's expense recognition policy is conservative or not. The analyst should also recognize that it is possible that two companies in the same industry have very different expense recognition policies.

4. Non-Recurring Items

Analysts are generally trying to estimate and assess future earnings of a company. Hence, reporting standards require firms to separate income and expense items that are likely to continue in the future, from items that are not likely to continue. (You will be more interested in items that will continue as compared to one-time items.)

Unusual or Infrequent Items

Both IFRS and US GAAP allow recognition of items that are unusual or infrequent. These items are also called exceptional items i.e., items not "inherent" to the company's current activities. Examples include restructuring charges and gains/losses from sale of equipment, receipts from a legal case, costs of integrating an acquisition, and impairment of intangible

assets, etc. These items are shown as part of a company's continuing operations but are presented separately.

While forecasting future earnings, an analyst should assess whether the items reported as unusual or infrequent are likely to reoccur. Analysts should not simply ignore all unusual items.

Discontinued Operations

A discontinued operation is an operation which a company has disposed of or plans to dispose of. Net income from discontinued operations is shown (as a separate line item on the income statement) net of tax after net income from continuing operations.

Assets and liabilities related to the discontinued operations are aggregated and recognized on the balance sheet as held for sale.

Since the discontinued operation will no longer provide earnings to the company, an analyst may exclude discontinued operations when forecasting future earnings.

Changes in Accounting Policy

At times, new accounting standards may require companies to change accounting policies. An example can be changing the inventory valuation method from last in, first out (LIFO) to first in, first out (FIFO). Companies are allowed to adopt standards prospectively (in the future) or retrospectively.

- Retrospective application means that the financial statements for previous years are presented as if the newly adopted accounting principle had been used throughout the period. A change in accounting policy is applied retrospectively. For example, if a company shifts from LIFO valuation method to FIFO valuation method, this change will require a retrospective application.
- Prospective application means that only the financial statements for the period of change and for future periods are changed. Financial statements for previous years are not changed. At times, new standards might require companies to change accounting estimates such as the useful life of a depreciable asset. Changes in accounting estimates are applied prospectively.
- Correction of an error for a prior period is another possible adjustment which requires a restatement of the four major financial statements. If a company is making corrections very often, this gives a negative signal and investors will avoid investing in such a company.

Modified retrospective approach: According to new revenue recognition standards, companies can also use "modified retrospective" method of adoption. Under this approach, companies can adjust opening balances of retained earnings (and other applicable accounts) for the cumulative impact of the new standard. They are not required to revise previously reported financial statements.

5. Earnings Per Share

Simple versus Complex Capital Structure

Earnings per share (EPS) is a very important profitability measure. It depicts the earnings per ordinary share. Some basic terminologies related to EPS are:

- Potentially dilutive securities: Securities that can be converted into ordinary shares are called potentially dilutive securities. This includes convertible bonds, convertible preferred stock, and employee stock options.
- Simple capital structure: If a company has no potentially dilutive securities it is said to have a simple capital structure.
- Complex capital structure: If a company has potentially dilutive securities it is said to have a complex capital structure.
- Dilutive securities: A potentially dilutive security that decreases EPS when exercised is called a dilutive security.
- Antidilutive security: A potentially dilutive security that increases EPS when exercised is called an antidilutive security.

Basic EPS

Basic EPS is the amount of income available to common shareholders divided by the weighted average number of common shares outstanding over a period. Basic EPS is calculated as:

$$\text{Basic EPS} = \frac{\text{Net income} - \text{Preferred dividends}}{\text{Weighted average number of shares outstanding}}$$

In this calculation we do not consider the effect of any potentially dilutive securities.

Weighted average number of shares outstanding is the number of shares outstanding during the year, weighted by the portion of the year they were outstanding. Stock splits and stock dividends are applied retroactively to the beginning of the year, so the old shares are converted to new shares for consistency.

Example

During 2018, Company ABC had a net income of \$100,000. It paid \$22,000 as dividends to its preference shareholders and \$12,000 as dividends to its common shareholders. The number of common shares outstanding during 2018 was as follows:

Shares as of January 1, 2018:	10,000
Additional shares issue on July 1, 2018:	2,000

Calculate the basic EPS of the company for 2018.

Solution:

We had 10,000 shares outstanding for the first 6 months and 12,000 shares outstanding for

the last 6 months.

Therefore, weighted average number of shares outstanding = $10,000 \times 6/12 + 12,000 \times 6/12 = 11,000$ shares.

$$\text{Basic EPS} = \frac{\$100,000 - \$22,000}{11,000} = \$7.09$$

Note: We ignore dividend paid to common shareholders.

Diluted EPS: The If-Converted Method

In this calculation, we consider the effect of potentially dilutive securities. If a firm has a complex capital structure it has to report both basic and diluted EPS. Diluted EPS is calculated as:

$$\text{Diluted EPS} = \frac{\text{Net Income} + \text{After tax interest} - \text{Preferred dividend} + \text{convertible preferred dividends}}{\text{Weighted Average Shares} + \text{New shares if convertible debt is converted}}$$

For 'preference shares', we need to subtract preference share dividends from the numerator and add new shares issued from conversion to the denominator.

Example

During 2018, Company ABC had a net income of \$100,000. It paid \$22,000 dividends to its preference shareholders and \$12,000 dividends to its common shareholders. It had 2,200 preference share and 11,000 common shares outstanding during 2018. Each preference share is convertible into 2 shares of common stock. Calculate the diluted EPS for the company.

Solution:

Number of common shares issued upon conversion = $2,200 \times 2 = 4,400$

$$\text{Diluted EPS} = \frac{\text{NI} + \text{conv debt int} (1 - t) - \text{pref div} + \text{conv pref div}}{\text{Wt avg shares} + \text{New shares issued}}$$

$$\text{Diluted EPS} = \frac{\$100,000 + 0 - \$22,000 + \$22,000}{11,000 + 4,400} = \$6.5$$

For 'convertible bonds', we need to add the after-tax interest cost savings to the numerator and new shares issued from conversion to the denominator.

Example

During 2018, Company ABC had a net income of \$100,000. The capital structure of the company for 2018 was as follows:

11,000 common shares

1,000 convertible bonds with par value of \$100 and 10% coupon; convertible to 5,000 shares

The tax rate of the company is 30%.

Calculate diluted EPS.

Solution:

Number of common shares issued upon conversion = 5,000

Interest payable on the bonds = $100 \times \$1,000 \times 10\% = \$10,000$

$$\text{Diluted EPS} = \frac{\text{NI} + \text{conv debt int} (1 - t) - \text{pref div} + \text{conv pref div}}{\text{Wt avg shares} + \text{New shares issued}}$$

$$\text{Diluted EPS} = \frac{\$100,000 + \$10,000 \times 0.7 - \$0 + \$0}{11,000 + 5,000} = \$6.69$$

For 'stock options', we use the 'Treasury Stock Method', which assumes that the hypothetical funds received by the company from the exercise of options are used to purchase shares of the company's common stock at the average market price over the reporting period. Thus, the numerator is unchanged and the number of shares to be added to the denominator = the number of shares created by exercising the options - number of shares hypothetically repurchased with the proceeds of the exercise.

Example

During 2018, Company ABC had a net income of \$100,000. It paid \$22,000 dividends to its preference shareholders and \$12,000 dividends to its common shareholders. The capital structure of the company for 2018 was as follows:

11,000 common shares

1,000 stock options outstanding, that have an exercise price of \$20.

During 2018, the average market price for the company's share was \$25.

Calculate the diluted EPS.

Solution:

Number of common shares issued upon conversion = 1,000

Cash proceeds from the exercise of options = $1,000 \times 20 = \$20,000$

Number of shares that can be purchased at the average market price with these funds = $\$20,000 / 25 = 800$

Net increase in common shares outstanding = $1,000 - 800 = 200$

$$\text{Diluted EPS} = \frac{\text{NI} + \text{conv debt int} (1 - t) - \text{pref div} + \text{conv pref div}}{\text{Wt avg shares} + \text{New shares issued}}$$

$$\text{Diluted EPS} = \frac{\$100,000 + \$0 - \$22,000 + \$0}{11,000 + 200} = \$6.96$$

Other Issues with Diluted EPS and Changes in EPS

Some potentially convertible securities could be antidilutive. Including them in the calculations would result in an EPS that is higher than the company's basic EPS. Such securities should not be included in the calculation of diluted EPS.

Instructor's Note:

Assess each instrument individually and determine if it is dilutive or not. Only instruments which are dilutive must be included in the diluted EPS calculation.

Changes in EPS

In general, an EPS can increase either due to an increase in net income, a decrease in the number of shares outstanding, or a combination of both.

6. Income Statement Ratios and Common-Size Analysis

Common-Size Analysis of the Income Statement

Common-size income statement presents each line item on the income statement as a percentage of revenue. This format standardizes the income statements and helps remove the effects of company size. They are useful to comparisons across time periods and across companies.

Income Statement Ratios

The income statement is used to calculate **income statement ratios** to evaluate a firm's profitability. The commonly used ratios are:

Gross profit margin = Gross profit / Revenue

Operating profit margin = Operating profit / Revenue

Net profit margin = Net profit / Revenue

High margin ratios are desirable. A firm can increase its margins by either increasing selling price or by lowering costs, or both.

An example of a common size income statement is shown below.

	2018	%	2017	%
Revenue	\$100,000	100%	\$110,000	100%
Cost of goods sold	\$60,000	60%	\$65,000	59%
Gross profit	\$40,000	40%	\$45,000	41%
SG&A	\$10,000	10%	\$11,000	10%
Depreciation expense	\$10,000	10%	\$11,000	10%
Operating profit	\$20,000	20%	\$23,000	21%
Interest expense	\$5,000	5%	\$5,500	5%
Earnings before taxes	\$15,000	15%	\$17,500	16%
Taxes (10%)	\$1,500	1.5%	\$1,750	1.6%
Net income	\$13,500	13.5%	\$15,750	14.3%

Looking at the above common-size statement, we can conclude that, the profitability margins of this company have declined in 2018 as compared to 2017.

Summary

LO: Describe general principles of revenue recognition, specific revenue recognition applications, and implications of revenue recognition choices for financial analysis.

Under the accrual method of accounting, revenue should be recognized when earned and not necessarily when cash is received.

The following five steps must be followed in order to recognize revenue:

1. Identify the contract(s) with a customer.
2. Identify the performance obligations in the contract.
3. Determine the transaction price.
4. Allocate the transaction price to the performance obligations in the contract.
5. Recognize revenue when (or as) the entity satisfies a performance obligation

Analysts can encounter many companies with complex revenue recognition policies such as:

- Principal versus agent
- Franchising/Licensing
- Software as a service or license
- Long-Term Contracts
- Bill and hold arrangements

Firms can use any revenue recognition technique provided there is a rationale behind their choice. Firms using an aggressive revenue recognition method will most likely inflate the earnings of the current period and report lower revenues in later periods. An analyst should consider the effects different revenue recognition methods can have on the financial statements of a company.

LO: Describe general principles of expense recognition, specific expense recognition applications, implications of expense recognition choices for financial analysis and contrast costs that are capitalized versus those that are expensed in the period in which they are incurred.

The most important principle of expense recognition is the matching principle, under which the expenses incurred to generate revenue are recognized in the same period as revenue.

Expenses that cannot be tied directly to generation of revenues are called periodic costs. They are expensed in the period incurred.

If an asset is expected to provide benefits only for the current period, its cost is expensed on the income statement for that period.

If an asset is expected to provide benefits over multiple periods, its cost is capitalized on the balance sheet and spread over the life of the asset.

If a company's policies result in early recognition of expenses, it can be considered a

conservative approach. On the other hand, if a company's policies delay the recognition of expenses, it can be considered an aggressive approach.

LO: Describe the financial reporting treatment and analysis of non-recurring items (including discontinued operations, unusual or infrequent items) and changes in accounting policies.

Net income from discontinued operations is shown net of tax after net income from continuing operations.

Both IFRS and U.S. GAAP allow recognition of unusual or infrequent items.

Changes in accounting policies can be adopted retrospectively (the financial statements for all fiscal years are presented as if the newly adopted accounting principle had been used throughout the period) or prospectively (only the financial statements for the period of change and for future periods are changed).

LO: Describe how earnings per share is calculated and calculate and interpret a company's basic and diluted earnings per share for companies with simple and complex capital structures including those with antidilutive securities.

When a company has simple capital structure, basic EPS is calculated using the formula:

$$\text{Basic EPS} = \frac{\text{Net Income} - \text{Preferred dividends}}{\text{Weighted Average Number of Shares Outstanding}}$$

When a company has complex capital structure, diluted EPS is calculated using the formula:

$$\text{Diluted EPS} = \frac{\text{Net Income} + \text{After tax interest} - \text{Preferred dividend} + \text{convertible preferred dividends}}{\text{Weighted Average Shares} + \text{New shares if convertible debt is converted}}$$

LO: Evaluate a company's financial performance using common-size income statements and financial ratios based on the income statement.

Common-size analysis of the income statement can be performed by stating each line item on the income statement as a percentage of revenue. Common-size statements facilitate comparison across time periods as well as across companies because the standardization of each line item removes the effect of size.

Net profit margin is calculated as: Net Income / Sales. This indicates how much income a company was able to generate for each dollar of revenue.

Gross profit margin is calculated as: Gross Profit / Sales. Where gross profit is calculated as revenue minus cost of goods sold.

Operating profit margin is calculated as: Operating Profit / Sales.

Analysts can use these profit margins to compare over time and with industry peers.